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BAA TECHNICAL NARRATIVE

CIEMS + AI Mechanic Unified Autonomy Assurance Capability

Tactical Unmanned Aircraft Systems (UAS) — Group 3-4 Autonomous Operations

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1.0 Executive Summary

The proliferation of autonomous decision-making capabilities aboard Group 3 and Group 4 Tactical Unmanned Aircraft Systems (UAS) has fundamentally altered the character of modern multi-domain operations. These platforms now execute intelligence, surveillance, and reconnaissance (ISR), strike-enablement, and electronic warfare functions with progressively reduced human involvement — operating beyond line of sight (BLOS), in GPS-degraded environments, and under adversarial electronic warfare (EW) conditions. As autonomy depth increases, so does the imperative for robust, auditable, and policy-compliant assurance infrastructure. Today, that infrastructure does not exist in a standardized, platform-agnostic form. This proposal addresses that gap directly.

The proposing team offers the **CIEMS + AI Mechanic Unified Autonomy Assurance Framework** — a dual-layer architecture designed to provide end-to-end autonomy assurance for Group 3–4 Tactical UAS platforms without requiring airframe modification. The framework operates as a safety-layer overlay integrating seamlessly with existing avionics, ground control station (GCS) architectures, and DoD data infrastructure.

At its core, the framework comprises two mutually reinforcing pillars. **AI Mechanic** is an onboard and ground-tethered runtime assurance subsystem that performs continuous inference monitoring, behavioral envelope enforcement, confidence score auditing, anomaly detection, and graduated autonomous intervention — all in real time during live flight operations. **CIEMS** (Continuous Integrated Engineering and Mission Support) is a persistent lifecycle engineering integrity layer that maintains model provenance, training lineage, configuration version control, supply chain integrity validation, and automated mission readiness certification records from model development through operational deployment and platform retirement.

Together, these pillars create an unbroken chain of custody from the moment an AI model is trained through every flight operation in which it executes decisions. The framework addresses four categories of mission-critical risk that currently lack standardized mitigation on DoD autonomous UAS platforms: AI behavioral drift under operational distribution shift; adversarial sensor and model manipulation; the absence of post-incident forensic capability; and the lack of auditable documentation supporting DoD AI Ethics Principles and **DoDD 3000.09** compliance.

The proposed effort spans 36 months across three phases, advancing from laboratory integration and hardware-in-the-loop (HWIL) testing (Phase 1, TRL 4) through platform integration and limited flight demonstration (Phase 2, TRL 5-6) to operational evaluation and airworthiness coordination (Phase 3). Total notional cost is \$20.5 million. The framework is designed for insertion into existing DoD programs of record with minimum integration friction, yielding a path to scalable, policy-compliant autonomous UAS operations at operationally relevant tempo.

KEY PROPOSAL CLAIMS

- Platform-agnostic autonomy assurance overlay for Group 3-4 UAS — no airframe modification required
 - Dual-pillar architecture: runtime monitoring (AI Mechanic) + lifecycle integrity (CIEMS)
 - End-to-end chain of custody from model training through live autonomous flight
 - >97% behavioral anomaly detection within 200ms; forensic reconstruction at $\geq 10\text{Hz}$ resolution
 - Full compliance architecture for DoDD 3000.09, DoD AI Ethics Principles, NIST AI RMF, and MIL-STD-882E
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2.0 Mission Context and Operational Need

2.1 Tactical UAS Operating Environment

Group 3 and Group 4 UAS platforms — defined by the DoD UAS Task Force taxonomy as systems ranging from 55 to 1,320 pounds maximum gross takeoff weight (MGTOG), operating at altitudes up to 18,000 feet mean sea level (MSL) — constitute the tactical backbone of current and projected multi-domain operations. These platforms execute a growing portfolio of AI-governed mission functions including persistent ISR, target geolocation, moving-target indication (MTI), communications relay, and increasingly, strike-enablement in support of Joint All-Domain Command and Control (JADC2) architectures.

The operational environments in which these platforms operate are characterized by multiple compounding degradation factors. **GPS jamming** and spoofing by near-peer adversaries eliminates reliable positioning for navigation and weapons employment. **Communications latency** and denial in contested electromagnetic environments render continuous human-in-the-loop control operationally infeasible, necessitating true autonomous execution for defined mission segments. **Electronic warfare (EW) threats** including radar jamming, datalink disruption, and cognitive electronic attack degrade sensor fidelity and command authority simultaneously.

Autonomous flight functions now resident on these platforms include waypoint navigation with dynamic re-routing, terrain-following and obstacle avoidance, electro-optical/infrared (EO/IR) target identification and tracking, **Ground-Based Sense and Avoid (GBSAA)** integration, and — in some configurations — rules-of-engagement (ROE) boundary enforcement as an AI-governed function. Each of these functions involves AI inference operating without real-time human supervision during the decision execution window. The integrity of those decisions is not currently subject to standardized runtime assurance.

2.2 The Autonomy Assurance Gap

Despite the increasing operational reliance on AI-governed autonomous functions, the DoD currently lacks a standardized, platform-agnostic framework for runtime AI behavioral monitoring, post-incident forensic reconstruction, and adversarial failure-mode coverage across tactical UAS platforms. This gap manifests across three distinct dimensions.

Policy Compliance Dimension: DoDD 3000.09 (Autonomous Weapons Systems, updated 2023) requires that autonomous and semi-autonomous weapons systems be designed to allow commanders and operators to exercise appropriate levels of human judgment over the use of force. It further requires that systems fail safely and predictably. These requirements cannot be met without runtime monitoring infrastructure capable of detecting behavioral deviations and executing graduated intervention protocols. Similarly, the DoD's five **AI Ethics Principles** — responsible, equitable, traceable, reliable, and governable — require technical implementation mechanisms that do not exist in a standardized form on current tactical UAS platforms. Traceability and governability, in particular, require the kind of persistent logging and lifecycle documentation that CIEMS provides.

Mission Readiness Dimension: Without runtime behavioral monitoring, program managers and airworthiness authorities lack the data necessary to characterize the behavioral envelope of deployed AI systems under operational conditions. This creates a latent risk of undetected behavioral drift — a condition in which an AI model's decision outputs gradually diverge from validated behavior as operational conditions deviate from the training distribution. This drift is not detectable through pre-mission testing alone; it manifests in the live operational environment and requires continuous monitoring to identify.

Accountability Dimension: Post-incident investigation of autonomous system behaviors — including potential fratricide, civilian harm, or mission failure attributable to AI malfunction — currently requires reconstruction from fragmentary data sources with no standardized forensic fidelity standard. Investigation timelines measured in weeks or months are operationally and politically untenable. The absence of tamper-evident, high-fidelity inference logging constitutes both an accountability gap and an acquisition liability for program offices fielding autonomous systems.

MISSION READINESS RISK STATEMENT

The absence of standardized autonomy assurance infrastructure on deployed Group 3-4 UAS platforms constitutes a mission readiness risk and a DoD policy compliance gap. Undetected behavioral drift, adversarial manipulation, and the lack of forensic accountability represent unmitigated hazards under current

operational configurations.

2.3 Adversarial Failure Mode Landscape

The following failure modes define the threat surface against which the CIEMS + AI Mechanic framework is designed to provide coverage. These modes are drawn from open-source AI safety literature, DoD red team findings in the public domain, and established adversarial machine learning taxonomy.

Failure Mode	Description	Operational Impact
AI Behavioral Drift	Gradual divergence of AI decision outputs from validated envelope as operational environmental conditions deviate from training distribution (distribution shift)	Undetected mission-critical decision errors; ROE boundary violations; navigation failures
Adversarial Sensor Spoofing	Manipulation of GPS, EO/IR, or radar altimeter sensor inputs to induce incorrect AI inference — including false target identification or navigation waypoint corruption	Weapons employment error; platform loss; fratricide risk
Model Poisoning via Supply Chain Compromise	Unauthorized modification of AI model weights or training data during the development or integration pipeline, producing malicious behavior triggered under specific operational conditions	Latent adversary-controlled AI behavior; mission compromise
Decision Boundary Erosion	Degradation of AI classifier decision boundaries under sustained high-tempo operations, producing increasing error rates that are not captured by pre-mission validation	Progressive mission performance degradation; undetected accuracy loss

Failure Mode	Description	Operational Impact
Confidence Score Manipulation	Adversarial inputs designed to produce artificially high model confidence scores on incorrect classifications, particularly in target identification functions	False-positive target identification; unauthorized lethal engagement
Cascade Failures in Swarm Coordination	Propagation of a single agent's behavioral anomaly through multi-agent UAS swarm coordination logic, amplifying the effect of a localized failure across the entire swarm	Mission-wide autonomous system failure; loss of multiple platforms

3.0 Technical Approach

3.1 Architecture Overview: CIEMS + AI Mechanic Unified Framework

The proposed capability is organized as a dual-pillar unified framework, depicted conceptually below. Both pillars share a common integration bus and data model, enabling seamless information flow between runtime operations and lifecycle engineering records. This integration eliminates the data silos that currently separate operational flight data from engineering configuration management — a separation that is a root cause of the autonomy accountability gap described in Section 2.2.

Pillar 1 — AI Mechanic (Runtime Assurance Layer): An onboard and ground-tethered monitoring subsystem that performs continuous inference monitoring, behavioral envelope checking, confidence score auditing, and anomaly flagging during live flight operations. AI Mechanic operates at the edge — executing on embedded compute co-located with the AI inference engine — and at the GCS, where operator-facing dashboards surface autonomy health status and **Autonomy Caution Events (ACEs)** in real time.

Pillar 2 — CIEMS (Lifecycle Engineering Integrity Layer): A persistent data fabric that maintains model provenance, training lineage, configuration version

control, and workflow integrity records from development through deployment and platform retirement. CIEMS serves as the authoritative system of record for all engineering activities affecting autonomous system software and hardware configurations, providing the evidentiary foundation for airworthiness certification, DoD AI policy compliance, and post-incident investigation.

Both pillars are integrated through the **Autonomy State Bus (ASB)**, a lightweight publish-subscribe middleware layer compatible with ROS2/DDS messaging protocols and **STANAG 4586** UAS control and non-payload communications (CUCS) data interfaces. The ASB enables real-time state sharing between the runtime monitoring functions of AI Mechanic and the configuration integrity records maintained by CIEMS, creating a closed-loop assurance architecture in which runtime anomalies automatically trigger CIEMS audit queries to determine whether detected deviations are attributable to configuration changes, model updates, or environmental factors.

Framework Layer	Pillar	Primary Domain	Key Functions	Execution Locus
Runtime Assurance	AI Mechanic	Live flight operations	Behavioral monitoring, ACE detection, forensic logging, intervention	Edge compute (onboard) + GCS
Lifecycle Integrity	CIEMS	Engineering & acquisition	Model provenance, config control, SBOM validation, MRR generation	Classified cloud (IL4/IL5) + GCS
Integration Bus	Autonomy State Bus (ASB)	Cross-pillar data fabric	Real-time state sharing, event correlation, anomaly-to-audit linkage	Distributed (onboard + ground)

3.2 Runtime Assurance Functions (AI Mechanic)

AI Mechanic implements five core runtime assurance functions, each addressing a distinct category of autonomous system risk. All functions operate continuously

during AI-governed flight segments and are configurable per mission profile through pre-mission parameter sets loaded by CIEMS at mission initialization.

3.2.1 Behavioral Envelope Monitor (BEM)

The **Behavioral Envelope Monitor (BEM)** continuously compares live AI decision outputs — including classification outputs, navigation commands, and engagement authorization signals — against a certified behavioral envelope established during pre-mission validation. The behavioral envelope is defined as a multidimensional bounded space of acceptable AI outputs across the full range of expected mission scenarios, validated through hardware-in-the-loop (HWIL) testing prior to each mission class. Deviations from the envelope exceeding configurable thresholds are flagged as **Autonomy Caution Events (ACEs)** and routed to the Autonomous Intervention Controller for graduated response. The BEM operates at a minimum evaluation rate of 10Hz, ensuring sub-100ms detection latency for sustained envelope violations.

3.2.2 Drift Detection Engine (DDE)

The **Drift Detection Engine (DDE)** applies statistical process control (SPC) methods — specifically **Cumulative Sum (CUSUM)** and **Exponentially Weighted Moving Average (EWMA)** algorithms — to time-series streams of AI inference outputs, detecting gradual behavioral drift that may not produce immediate envelope violations but represents a trend toward degraded or anomalous performance. The DDE is particularly critical for long-duration ISR missions in which distribution shift may accumulate gradually due to changing environmental conditions, time-of-day lighting variation, or adversarial environmental manipulation. Drift alerts are generated when accumulated drift exceeds mission-specific tolerance bands defined in the pre-mission configuration.

3.2.3 Confidence Score Auditor (CSA)

The **Confidence Score Auditor (CSA)** monitors model confidence outputs in real time, applying statistical anomaly detection to identify implausible confidence patterns that may indicate adversarial manipulation, sensor spoofing, or model degradation. The CSA maintains a running statistical model of expected confidence distributions for each mission function and flags deviations — including abnormally high confidence scores on novel inputs (a hallmark of adversarial examples) and systematic confidence suppression (indicative of model poisoning or input

manipulation). CSA outputs are correlated with BEM findings to reduce false positive rates and prioritize intervention actions.

3.2.4 Forensic Replay Module (FRM)

The **Forensic Replay Module (FRM)** maintains a tamper-evident, time-stamped log of all AI inference events, sensor inputs, decision outputs, and system state data in a write-once ring buffer architecture. All log entries are cryptographically signed using platform-resident keys managed by the CIEMS Model Provenance Ledger, ensuring integrity for post-incident evidentiary use. The FRM provides sub-second temporal resolution at a minimum logging rate of 10Hz, enabling complete reconstruction of autonomous decision sequences for post-incident investigation. Storage capacity is sized to retain a minimum of 24 hours of continuous flight data at full logging resolution, with configurable overflow behavior (oldest-data overwrite or alert-triggered secure dump to GCS).

3.2.5 Autonomous Intervention Controller (AIC)

The **Autonomous Intervention Controller (AIC)** is the executive function of AI Mechanic, receiving ACE inputs from BEM, DDE, and CSA and executing a pre-approved graduated response ladder. The response ladder is as follows:

1. **Level 1 — Alert:** GCS operator notification within 500ms of ACE detection.
No system action; operator retains full authority.
2. **Level 2 — Watchdog:** AI Mechanic activates a secondary validation check on AI outputs. Mission continues with enhanced monitoring. Operator notified of elevated monitoring state.
3. **Level 3 — Graceful Degradation:** Specific AI function flagged as unreliable is disabled or constrained to a reduced operational envelope. Alternative non-AI guidance modes activated where available.
4. **Level 4 — Safe-State Handover:** Upon confirmed critical ACE (e.g., sustained envelope violation, confidence manipulation, or loss of BEM function), AIC executes platform safe-state entry — transitioning to a pre-programmed loiter profile or RTB sequence — within 2 seconds of trigger, regardless of communications status. Human operator is notified and can override safe-state entry during the first 5-second window.

3.3 Engineering Lifecycle Integrity Functions (CIEMS)

CIEMS provides the persistent engineering integrity backbone of the unified framework. Where AI Mechanic addresses what the AI system is doing during flight, CIEMS addresses what the AI system is and how it got there — from model conception through operational retirement.

3.3.1 Model Provenance Ledger (MPL)

The **Model Provenance Ledger (MPL)** maintains a cryptographically signed, append-only record of each AI model's complete development lineage. Each MPL entry captures training dataset provenance and version, hyperparameter configurations, validation and verification (V&V) test results, approval chain documentation (including program office, airworthiness authority, and test evaluation sign-offs), and the cryptographic hash of the approved model binary. The MPL serves as the authoritative chain of custody record for all AI models operating on the platform, and is the primary evidentiary artifact for DoD AI policy compliance audits and post-incident investigation.

3.3.2 Configuration Version Control (CVC)

The **Configuration Version Control (CVC)** subsystem implements Git-based versioning of all mission-critical software components, with cryptographically hash-verified deployment manifests that establish an unambiguous record of the exact software configuration present on the platform at any given mission. CVC prevents unauthorized modification through mandatory hash verification at mission initialization — any discrepancy between the deployed component hash and the CIEMS-recorded approved hash halts mission execution and generates a configuration integrity alert requiring resolution by an authorized engineering authority before proceeding.

3.3.3 Workflow Integrity Monitor (WIM)

The **Workflow Integrity Monitor (WIM)** enforces sequencing and authorization requirements across the engineering lifecycle, detecting and alerting on out-of-sequence process execution, unauthorized component substitutions, and deviations from approved maintenance and update procedures that could compromise system safety. WIM integrates with program office workflow management systems to enforce the engineering change control process as a technical constraint rather than relying solely on procedural compliance.

3.3.4 Supply Chain Integrity Validator (SCIV)

The **Supply Chain Integrity Validator (SCIV)** compares the cryptographic signatures of all deployed hardware and software components against a trusted **Hardware and Software Bill of Materials (HSBOM/SBOM)** maintained in CIEMS. SCIV validation is executed at mission initialization and after any maintenance action, flagging any discrepancy between the deployed configuration and the approved bill of materials as a potential supply chain integrity event requiring engineering authority disposition before mission execution is authorized. SCIV is designed in compliance with Executive Order 14028 (Improving the Nation's Cybersecurity) SBOM requirements and applicable NIST guidance on software supply chain security.

3.3.5 Mission Readiness Report (MRR)

The **Mission Readiness Report (MRR)** is an automated pre-mission certification artifact generated by CIEMS, summarizing the results of all integrity checks — CVC hash verification, SCIV SBOM validation, MPL approval status, and WIM workflow compliance — along with any open anomalies, deviation dispositions, and waiver status. The MRR is generated within 90 seconds of pre-mission initialization and provides the authoritative pre-flight autonomy airworthiness documentation required by airworthiness authorities. The MRR is cryptographically signed and archived in CIEMS, creating a complete pre-flight certification record for every mission execution.

3.4 Integration Architecture

The CIEMS + AI Mechanic framework is engineered as a platform-agnostic overlay, requiring no airframe structural modification and imposing minimal integration burden on existing avionics architectures. Integration is accomplished through the **CIEMS Integration Module (CIM)**, a hardware/software package sized for installation in a standard payload bay or ground control station rack unit.

Platform Integration: The CIM is designed for compatibility with representative Group 3-4 airframes including the MQ-1C Gray Eagle, RQ-7B Shadow, and Group 4 persistent ISR variants. Integration leverages existing avionics data bus access points, requiring no new airframe penetrations. The CIM interfaces with onboard avionics via **MIL-STD-1553B** legacy data bus for existing avionics parameters, with supplemental Ethernet (1000BASE-T) connectivity for high-bandwidth AI inference data streams.

Ground Control Station (GCS) Integration: AI Mechanic provides a dedicated operator display panel embedded in the GCS operator interface, presenting real-time autonomy health indicators, active ACE alerts, drift trend indicators, and human override initiation controls. The display is compliant with **ARINC 661** display interface standards to ensure compatibility with existing GCS display management systems.

Cloud/Edge Hybrid Architecture: CIEMS backend services operate on classified cloud infrastructure certified at **DoD Impact Level 4/5 (IL4/IL5)**, providing the storage, processing, and access control capabilities required for the MPL, CVC, WIM, and MRR functions. AI Mechanic runtime functions (BEM, DDE, CSA, FRM, AIC) execute on edge compute hardware co-located with the AI inference engine — an NVIDIA Jetson Orin-class system-on-module or equivalent — providing the low-latency processing required for sub-200ms anomaly detection.

Data Interface Standards: The framework implements the following standard interfaces to maximize interoperability and minimize platform-specific integration work:

Standard	Application	Integration Point
STANAG 4586	UAS interoperability — control and non-payload communications (CUCS)	ASB to GCS and UAS control network
ARINC 661	Display interface standard for GCS operator panels	AI Mechanic GCS display
MIL-STD-1553B	Legacy avionics data bus — navigation, sensor, and platform health data	CIM to platform avionics
REST / gRPC APIs	Cloud service integration for CIEMS backend functions	GCS to CIEMS IL4/IL5 cloud
ROS2 / DDS	Autonomy State Bus middleware for intra-framework messaging	AI Mechanic to CIEMS ASB

3.5 Autonomy Level Alignment

The framework is designed to operate across all relevant DoD autonomy modes as defined by **DoDD 3000.09** and characterized within the **NIST AI Risk Management Framework (AI RMF, AI 100-1)**. Capability behavior is tailored to the autonomy

mode declared by the mission authority pre-flight, ensuring appropriate human authority is preserved at every autonomy level.

Autonomy Mode	Human Authority	AI Mechanic Behavior	CIEMS Behavior
Human-on-the-Loop (HOL)	Operator retains all decision authority; AI provides recommendations only	Advisory ACE alerts; no automated intervention; all anomalies logged to FRM	Continuous MRR updates; all AI recommendations logged to MPL audit trail
Human-Supervised Autonomous	AI executes within pre-approved behavioral envelope; operator monitors and can intervene	Full BEM/DDE/CSA enforcement; ACE alerts and graduated AIC response Levels 1-3 active	Real-time configuration integrity monitoring; envelope deviation records archived
Fail-Safe Autonomous	Communications-denied or link-degraded; AI executes per pre-approved contingency plan	AIC Level 4 safe-state handover on confirmed critical ACE regardless of comm status; FRM continues logging	Post-mission MPL and FRM reconciliation upon link restoration; full mission record archived

4.0 Governance Model and Compliance Framework

4.1 AI Governance Architecture

The CIEMS + AI Mechanic framework instantiates a three-tier governance model that distributes AI authority across mission, engineering, and runtime domains — reflecting the distinct roles of combatant command, program office, and operational system in DoD AI governance. This model is designed to implement DoD AI Ethics Principles in technical form rather than relying on policy compliance alone.

Tier 1 — Mission Authority: The program office or designated combatant command authority defines behavioral envelopes, mission parameters, and ROE boundaries encoded into the system during the pre-mission configuration phase. Tier 1 authority is exercised through the CIEMS Mission Readiness Report workflow, which requires

authorized sign-off before mission parameters are loaded to the platform. Tier 1 decisions are cryptographically recorded in the CIEMS MPL and are not modifiable by Tier 2 or Tier 3 functions.

Tier 2 — Engineering Authority: The program engineering organization, operating through CIEMS, enforces software configuration control, manages deviation authority, and maintains the Model Provenance Ledger as the authoritative system of record. Tier 2 authority governs all changes to AI models, configuration parameters, and mission software components, ensuring that no unauthorized modification can reach the platform without engineering authority disposition.

Tier 3 — Runtime Authority: AI Mechanic enforces runtime behavioral constraints established by Tier 1 and validated by Tier 2. Tier 3 functions — BEM, DDE, CSA, and AIC — operate as a technical enforcement layer, not a decision authority. AIC intervention actions are pre-approved by Tier 1 and Tier 2 authorities through the MRR process; Tier 3 executes but does not decide the response rules.

4.2 DoD Policy Alignment

The framework is designed from inception for compliance with applicable DoD AI governance policy and technical standards. The following table maps each policy instrument to the specific framework components that implement its requirements:

Policy / Standard	Key Requirement	Implementing Framework Component
DoDD 3000.09 (Autonomous Weapons Systems, 2023)	Human-on-the-loop authority; predictable fail-safe behavior; documented override capability	AIC graduated response ladder; HOL mode; Tier 1/2/3 governance model; FRM audit trail
DoD AI Ethics Principles (2019)	Responsible, equitable, traceable, reliable, and governable AI	MPL (traceable); BEM/DDE/CSA (reliable); Three-tier governance (governable); FRM (accountable)
NIST AI RMF (AI 100-1)	Govern, Map, Measure, Manage core functions	Govern: Tier 1/2/3; Map: SCIV/MPL; Measure: BEM/DDE/CSA; Manage: AIC/WIM
MIL-STD-882E (System Safety)	Hazard identification, risk assessment, and	Behavioral envelope enforcement (hazard

Policy / Standard	Key Requirement	Implementing Framework Component
	mitigation	mitigation); AIC safe-state (risk reduction); BEM/DDE (detection)
JSIG / DoD RMF	Cybersecurity posture; DISA STIG compliance for embedded systems	CIM hardware hardening; CIEMS IL4/IL5 cloud posture; CVC hash verification; SCIV SBOM
EO 14028 (Cybersecurity, 2021)	Software supply chain security; SBOM requirements	SCIV SBOM validation; CVC deployment hash verification; MPL supply chain records

4.3 Airworthiness and Certification Pathway

The framework is designed for incremental airworthiness certification aligned with existing DoD airworthiness authority processes. The certification strategy addresses the novel challenge of certifying AI/ML components in a safety-critical aviation context, for which established regulatory frameworks provide only partial guidance.

Applicable Standards: Certification compliance is planned against **MIL-HDBK-516C** airworthiness criteria for software-intensive systems, with **DO-178C** software development assurance considerations applied to AI/ML components at the Design Assurance Level (DAL) appropriate to their hazard classification. The proposing team will coordinate with Army AMCOM and Air Force AFLCMC airworthiness authorities to define the specific DAL assignments and certification basis for each AI Mechanic and CIEMS component.

Incremental Certification Strategy:

5. **TRL 4 — Benchtop Validation:** Component-level V&V against defined behavioral envelope specifications; CIEMS MPL seeded with initial model provenance records; MRR prototype demonstrated.
6. **TRL 5 — Hardware-in-the-Loop (HWIL) Testing:** Full framework integration on representative platform avionics testbed; ACE detection performance against target metrics; airworthiness authority familiarization and preliminary certification basis agreement.

7. **TRL 6 — Flight Test Demonstration:** Instrumented flight test on designated Group 3-4 testbed aircraft; airworthiness documentation package submitted; limited operational evaluation under program office oversight.

CERTIFICATION PATHWAY NOTE

The proposing team acknowledges that AI/ML component certification in the DoD aviation context remains an evolving regulatory domain. The CIEMS MRR, MPL, and FRM functions are specifically designed to generate the documentation artifacts required by airworthiness authorities regardless of the specific certification standard ultimately applied — providing a certification-agnostic documentation foundation that can be mapped to applicable requirements as the regulatory landscape matures.

5.0 Operational Benefits and Performance Metrics

5.1 Quantified Capability Improvements

The following performance metrics represent design targets to be validated through Phase 1 HWIL testing and Phase 2 flight demonstration. Metrics are derived from operational analysis of the target mission scenarios and reflect the performance thresholds required to provide operationally meaningful autonomy assurance at Group 3-4 UAS operational tempo.

Performance Metric	Target Value	Measurement Method	Implementing Function
Autonomy Anomaly Detection Rate	>97% detection of behavioral envelope violations	Scripted anomaly injection against standard mission scenario library during HWIL testing	BEM, DDE, CSA

Performance Metric	Target Value	Measurement Method	Implementing Function
Anomaly Detection Latency	<200ms from onset of envelope violation	Instrumented timing measurement during HWIL anomaly injection	BEM
False Positive Rate	<2% across standard mission scenario library	False ACE count during normal mission profile execution in HWIL	BEM, DDE, CSA
Forensic Reconstruction Fidelity	100% of inference events captured at ≥ 10 Hz logging rate	FRM log completeness audit against independent inference event recorder	FRM
SBOM Integrity Coverage	100% of deployed software and hardware components covered	SCIV SBOM coverage audit at mission initialization	SCIV
Mission Readiness Certification Time	<90 seconds for automated MRR generation	Timed MRR generation across representative pre-mission scenarios	MRR / CIEMS
Operator Alert-to-Notification Latency	<500ms from ACE detection to GCS operator display	Instrumented end-to-end latency measurement from BEM trigger to GCS panel update	AIC, GCS display
Safe-State Handover Execution Time	<2 seconds from AIC Level 4 trigger to platform safe-state entry	Timed AIC Level 4 activation during HWIL and flight test	AIC

5.2 Mission Enablement Impact

Beyond measurable system performance metrics, the CIEMS + AI Mechanic framework delivers qualitative mission enablement benefits that address the most significant operational constraints currently limiting the expansion of autonomous UAS operations:

- Expanded BLOS Autonomous Operations:** The framework provides the behavioral assurance infrastructure that acquisition and operational

authorities require to authorize expanded BLOS autonomous mission execution with confidence in behavioral predictability and policy compliance.

- **Human-Machine Teaming at Operational Tempo:** By surfacing real-time autonomy health data to GCS operators in a clear, actionable format, AI Mechanic enables operators to maintain meaningful supervisory authority over autonomous functions at operationally relevant tempo — addressing the core human-on-the-loop intent of DoDD 3000.09 without degrading mission execution speed.
- **Post-Incident Investigation Acceleration:** The Forensic Replay Module reduces autonomous system post-incident investigation timelines from the current standard of weeks to hours, providing investigators with complete, tamper-evident reconstruction of all AI decisions and sensor inputs at sub-second temporal resolution.
- **Auditable AI Compliance Documentation:** CIEMS provides DoD acquisition authorities, program offices, and Inspector General functions with the auditable AI compliance documentation required to satisfy DoD AI Ethics Principles accountability requirements — enabling programs to demonstrate responsible AI deployment rather than asserting it.
- **Supply Chain Risk Reduction:** SCIV SBOM validation eliminates an entire class of software supply chain attack vectors by ensuring that only cryptographically verified, authority-approved software components are present on the platform at mission execution — a capability with growing significance as adversary focus on software supply chain compromise intensifies.

6.0 Work Breakdown Structure and Schedule

The proposed effort is structured in three sequential phases spanning 36 months, with clear TRL advancement milestones at each phase boundary. The Work Breakdown

Structure (WBS) below provides the programmatic framework for cost, schedule, and technical performance tracking.

WBS Element	Phase 1 (Mo. 1-12)	Phase 2 (Mo. 13-24)	Phase 3 (Mo. 25-36)
1.0 Systems Engineering & Architecture	Architecture definition; interface specifications; CONOPS development; TRL 4 validation plan	Architecture updates from lab findings; integration issue resolution; Phase 3 test planning	System architecture review; program-of-record insertion documentation
2.0 AI Mechanic Runtime Development	BEM, DDE, CSA, FRM, AIC software development; lab integration; HWIL test	Platform integration on testbed aircraft; GCS display integration; TRL 5-6 demonstration	Operational evaluation support; anomaly resolution; configuration finalization
3.0 CIEMS Integration & Lifecycle Tools	MPL, CVC, WIM, SCIV, MRR development; lab demonstration; cloud environment standup	Platform-specific CIEMS configuration; IL4/IL5 cloud migration; MRR automation testing	Operational CIEMS configuration; documentation package; transition support
4.0 Platform Integration & GCS	CIM hardware design; avionics interface specification; GCS display design	CIM fabrication and platform installation; GCS integration; ASB commissioning	Operational configuration refinement; integration issue closeout
5.0 Test and Evaluation	Lab V&V; HWIL testing with representative avionics; anomaly injection testing	Platform-level integration testing; limited flight demonstration (TRL 6)	Operational evaluation; independent T&E support; metrics validation
6.0 Certification & Documentation	Certification basis discussions; preliminary airworthiness documentation	Airworthiness authority coordination; DO-178C artifacts; MRR certification demonstration	Airworthiness documentation package; certification basis closeout; program-of-record submission
7.0 Program Management	Program plan; IMP/IMS; risk management; CDRLs; Government	Ongoing program management; risk/issue tracking; Government reporting	Final program reporting; lessons learned; transition package delivery

WBS Element	Phase 1 (Mo. 1-12)	Phase 2 (Mo. 13-24)	Phase 3 (Mo. 25-36)
	reporting		

Phase 1 (Months 1-12) — Architecture Refinement and Laboratory Integration: This phase focuses on detailed architecture definition, component-level software development for AI Mechanic and CIEMS functions, laboratory integration on a Government-furnished representative avionics testbed, and HWIL testing against the anomaly injection scenario library. Phase 1 concludes with a TRL 4 demonstration milestone reviewed by the program office and Government T&E authority.

Phase 2 (Months 13-24) — Platform Integration and Limited Flight Demonstration: CIM hardware fabrication and installation on the designated Group 3-4 testbed aircraft; GCS integration and AI Mechanic operator display commissioning; CIEMS IL4/IL5 cloud environment operational; limited flight demonstration under instrumented conditions with independent T&E observation. Phase 2 concludes with a TRL 5-6 demonstration milestone and preliminary airworthiness authority review.

Phase 3 (Months 25-36) — Operational Evaluation and Airworthiness Coordination: Extended operational evaluation under representative mission scenarios; independent T&E metrics validation; airworthiness documentation package completion; program-of-record insertion package development and delivery. Phase 3 concludes with a complete transition package suitable for program-of-record adoption.

7.0 Team Qualifications and Past Performance

The proposed team is structured to provide complementary expertise across the full technical and programmatic scope of this effort. The team composition reflects the multi-disciplinary requirements of autonomous systems assurance, integrating capability in embedded AI systems, DoD UAS platform integration, AI safety methodology, and independent test and evaluation.

Prime Contractor: The prime contractor brings demonstrated experience in autonomous systems software development, embedded AI inference systems, and DoD airworthiness compliance for software-intensive aviation systems. Past performance in analogous programs includes runtime monitoring systems for safety-critical autonomous platforms, AI/ML integration on DoD airborne systems, and GCS software development for Group 3-4 UAS. The prime contractor holds active Secret facility clearance and relevant ITAR registration.

Platform Integration Subcontractor: A subcontractor with direct experience in avionics integration on Group 3-4 UAS platforms, including hands-on familiarity with the CIM target platform families and established relationships with airworthiness authority technical liaisons. This team member provides the platform-specific integration expertise required to minimize integration risk during Phase 2 and maintains active involvement in DoD UAS interoperability working groups.

University Research Partner: A federally funded university research center with established programs in AI assurance, adversarial machine learning, and runtime verification for safety-critical systems. The research partner contributes the AI safety methodology foundation for BEM, DDE, and CSA — including CUSUM/EWMA algorithm adaptation for operational AI inference monitoring — and provides independent algorithm validation services. The research partner has active DoD research relationships and security-cleared personnel to support classified program elements.

FFRDC/UARC Independent T&E Support: A Federally Funded Research and Development Center (FFRDC) or University Affiliated Research Center (UARC) provides independent test and evaluation support for HWIL testing (Phase 1), flight demonstration (Phase 2), and operational evaluation (Phase 3). Independent T&E is essential to the credibility of TRL advancement claims and airworthiness documentation, and is structured to provide the program office with an unbiased assessment of framework performance against stated metrics.

The team has no organizational conflicts of interest with respect to the subject platforms and has established teaming agreements in place. All key personnel hold active security clearances appropriate to the program classification level. Classified facility support, if required for Phase 2 and 3 activities, is available through the prime contractor and platform integration subcontractor facility infrastructure.

8.0 Cost and Pricing Summary

The following cost structure represents a notional estimate developed for planning purposes in support of this BAA response. Final pricing will be submitted in response to a formal solicitation following any Government request for proposal. All costs are in then-year U.S. dollars.

Phase	Period	Primary Activities	Notional Cost	Recommended Contract Type
Phase 1	Months 1-12	Systems Engineering, Architecture, Laboratory Integration, HWIL Testing	\$4.2M	Cost-Plus-Fixed-Fee (CPFF)
Phase 2	Months 13-24	Platform Integration, GCS Integration, Limited Flight Demonstration	\$9.8M	Cost-Plus-Fixed-Fee (CPFF)
Phase 3	Months 25-36	Operational Evaluation, Airworthiness Coordination, Program-of-Record Package	\$6.5M	Firm-Fixed-Price (FFP)*
Total	36 Months	Full Program	\$20.5M	

* Phase 3 FFP is contingent upon demonstrated TRL 6 achievement at Phase 2 milestone review. If TRL 6 is not achieved at Phase 2 conclusion, Phase 3 will be executed under CPFF until TRL 6 is demonstrated, at which point remaining Phase 3 activities may transition to FFP by contract modification.

Cost Structure Rationale: CPFF is recommended for Phases 1 and 2 due to the inherent technical uncertainty associated with novel AI assurance system development, HWIL integration on Government-furnished avionics testbeds, and flight test operations. The cost-reimbursement vehicle appropriately allocates technical risk to the Government during the development and demonstration phases.

Phase 3 FFP is appropriate following TRL 6 demonstration, at which point the technical risk profile is substantially reduced and the remaining activities are primarily documentation, evaluation, and transition packaging — scope suitable for fixed-price execution.

Cost Sharing: The proposing team is prepared to discuss cost-sharing arrangements, including independent research and development (IRAD) contribution, as part of proposal negotiations. Specific cost-sharing proposals will be developed in response to a formal solicitation.

9.0 Data Rights and Intellectual Property

The proposing team is committed to a data rights posture that maximizes Government utility and supports competitive follow-on acquisition while providing appropriate protection for legitimate pre-existing proprietary investment.

Government Purpose Rights (GPR): All software, documentation, data, and technical information developed exclusively under this contract — including AI Mechanic runtime software, CIEMS integration software, the Autonomy State Bus middleware, integration documentation, and test data — will be delivered to the Government with **Government Purpose Rights (GPR)** as defined in DFARS 252.227-7013 and 252.227-7014. GPR delivery ensures that the Government may use, modify, reproduce, release, and disclose the developed items for any Government purpose, including competitive follow-on procurement, while restricting non-Government disclosure for the applicable GPR period.

Proprietary Rights Assertions: Assertions of proprietary rights (Limited Rights for technical data; Restricted Rights for computer software) will be made only for **pre-existing commercial intellectual property** developed entirely at private expense prior to this contract. All proprietary rights assertions will be specifically identified in the contract deliverable data list at contract award and will be subject to Government review and challenge per DFARS procedures. No proprietary rights will be asserted over any item developed with Government funding under this contract.

Open Architecture Commitment: The CIEMS + AI Mechanic framework is designed from the outset to comply with DoD open architecture principles, including compliance with the **Future Airborne Capability Environment (FACE)** technical

standard and **Modular Open Systems Approach (MOSA)** requirements where applicable. The Autonomy State Bus interfaces are documented in Government-deliverable interface control documents (ICDs) using open standard protocols (ROS2/DDS, STANAG 4586, REST/gRPC) to ensure that the framework does not create proprietary lock-in and that competitive follow-on development, integration of alternative AI functions, or third-party monitoring module integration is technically feasible without proprietary barriers.

Patent Rights: Inventions made under this contract will be subject to the Bayh-Dole Act (35 U.S.C. § 200 et seq.) and applicable DFARS patent rights clauses. The Government will retain a paid-up, nonexclusive, irrevocable license to practice or have practiced any subject invention for or on behalf of the United States. The proposing team will promptly disclose all subject inventions per contractual requirements.

Appendices

The following appendices are referenced within this technical narrative and will be included in full in the complete proposal submission package. Abbreviated reference descriptions are provided below for evaluator orientation.

Appendix	Title	Description
Appendix A	System Architecture Diagrams	Conceptual architecture block diagrams illustrating the CIEMS + AI Mechanic dual-pillar framework, Autonomy State Bus integration topology, CIM hardware block diagram, GCS display panel layout, and cloud/edge hybrid deployment architecture. Provided as reference only; does not represent a deliverable design baseline.
Appendix B	Adversarial Failure Mode Analysis Matrix	Structured analysis matrix mapping each adversarial failure mode

Appendix	Title	Description
		(Section 2.3) to applicable AI Mechanic detection functions, CIEMS prevention functions, AIC response actions, and residual risk assessment. Includes failure mode probability and severity ratings per MIL-STD-882E risk assessment methodology.
Appendix C	Compliance Cross-Reference Matrix	Detailed bidirectional compliance traceability matrix mapping DoDD 3000.09 paragraphs, NIST AI RMF sub-functions, MIL-STD-882E hazard categories, and DoD AI Ethics Principles to specific framework components (BEM, DDE, CSA, FRM, AIC, MPL, CVC, WIM, SCIV, MRR). Supports evaluator assessment of policy compliance completeness.
Appendix D	Acronym List	Complete alphabetical list of all acronyms and abbreviations used in this technical narrative, with full expansion and applicable reference standard or policy document where applicable.

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**BAA Technical Narrative —
CIEMS + AI Mechanic**